Yifan Feng

Ph.D. Candidate in Entomology — mosquito genomics, chromosomal inversions, and cytogenetics

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Education

PhD **Virginia Polytechnic Institute and State University**, Entomology — Blacksburg, VA, Aug 2021 – Dec 2026
- Advisor: Dr. Maria V. Sharakhova
- Dissertation: *Chromosomal Inversions Differentiate Mosquitoes in the Culex pipiens Complex*
- Department of Entomology and Fralin Life Sciences Institute
- Coursework: Molecular Genetics, Bioinformatics, Genomics, Biochemistry, Experimental Design

BS **Virginia Polytechnic Institute and State University**, Biological Sciences — Blacksburg, VA, Aug 2017 – May 2021

BS **Virginia Polytechnic Institute and State University**, Food Science and Technology — Blacksburg, VA, Aug 2017 – May 2021

Research Experience

Virginia Tech — Department of Entomology & Fralin Life Sciences Institute, Graduate Research Assistant — Blacksburg, VA, Aug 2021 – present
- Built **Hi-C libraries** and led the Hi-C component of **chromosome-scale mosquito genome assemblies** — trio-binning from **Nanopore long-read data**, **Hi-C scaffolding**, and **FISH** physical validation.
- Designed and validated **PCR breakpoint assays** to genotype chromosomal inversions across natural populations, linking structural variants to climatic adaptation and vector traits.
- Conducted population genomic analyses (**FST, PCA, AMOVA**) to quantify differentiation and gene flow among *Culex pipiens* populations across North America.
- Performed molecular dissections, DNA extraction, library preparation, and fluorescence microscopy imaging; maintained mosquito colonies.
- Mentored two undergraduate researchers in PCR optimization, experimental design, and scientific writing.
- Coordinated with collaborating groups across nine countries on the *Aedes aegypti* inversion study published in *Genome Biology and Evolution*.

Virginia Tech — Department of Food Science and Technology, Undergraduate Researcher — Blacksburg, VA, Sept 2020 – Dec 2020

Aerosolized *Listeria* contamination from cavitation treatment of raw produce.
- Applied microbubble cavitation treatments and quantified resulting aerosolized bacterial contamination.
- Built MATLAB models of cavitation effects and ran predictive simulations on university high-performance computing resources.
- Compiled, visualized, and statistically analyzed experimental results to identify contamination trends.

Molecular Genetics Laboratory, Research Intern | Blacksburg, VA | Mar 2020 – Dec 2021

- Performed DNA extraction, PCR amplification, and sequencing for mosquito population genetics studies.
- Managed sample inventory and quality control for field and laboratory collections.
- Gained hands-on experience with genome assembly workflows and variant calling pipelines.

Publications

Chromosomal Inversions and Their Potential Impact on the Evolution of Arboviral Vector *Aedes aegypti* — July 2025
Jiangtao Liang, Noah H. Rose, Ilya I. Brusentsov, Varvara Lukyanchikova, Dmitriy A. Karagodin, *Yifan Feng*, et al.
10.1093/gbe/evaf118 (Genome Biology and Evolution 17(7), evaf118)

Hi-C Proximity Ligation Approach Identified Chromosomal Rearrangements in *Culex pipiens* Mosquitoes — 2025
Yifan Feng, Varvara Lukyanchikova, Jiangtao Liang, Dmitriy A. Karagodin, Ilya I. Brusentsov, Megan L. Fritz, et al.
(American Journal of Tropical Medicine and Hygiene 112(6) — ASTMH Annual Meeting abstract)

Chromosomal Rearrangements Differentiate Species of the *Culex pipiens* Complex — 2025
Maria V. Sharakhova, *Yifan Feng*, Varvara Lukyanchikova, Jiangtao Liang, Dmitriy A. Karagodin, et al.
(Conference proceedings)

Chromosomal Inversions Differentiate Mosquitoes in the *Culex pipiens* Complex — 2026
Yifan Feng, Varvara Lukyanchikova, Jiangtao Liang, Ilya I. Brusentsov, Dmitriy A. Karagodin, Megan L. Fritz, Maria V. Sharakhova
(Manuscript in preparation)

Chromosome-Scale Genome Assembly of *Aedes albopictus* — 2026
Yifan Feng, et al.
(Manuscript in preparation)

Conference Presentations

- **Talk (accepted)** — *Hi-C proximity ligation reveals chromosomal inversions potentially associated with host preference in Culex pipiens mosquitoes.* Student 10-Minute Presentation Competition, Entomology 2026 (ESA Annual Meeting), Columbus, OH, November 2026.
- **Poster** — *Hi-C based discovery of structural variants in the Culex pipiens complex.* American Society of Tropical Medicine and Hygiene (ASTMH) Annual Meeting, November 2024.
- **Presentation** — *Chromosomal inversions differentiate mosquitoes in the Culex pipiens complex.* CeZAP Infectious Disease Symposium, Blacksburg, VA, October 2023.

Teaching and Mentoring

Virginia Tech — Department of Entomology, Graduate Teaching Assistant, ENT 2004 (Insects and Human Society) | Blacksburg, VA | Aug 2023 – May 2024

- Led weekly laboratory sessions and discussion sections for a general-education course serving non-biology majors.
- Taught insect taxonomy, physiology, and ecology exercises; graded laboratory reports and examinations.
- Developed visual teaching materials and provided one-on-one mentorship.

Sharakhova Lab, Virginia Tech, Research Mentor, Undergraduate Researchers — Blacksburg, VA — Jan 2024 – Dec 2025

- Mentored two undergraduate researchers through full project cycles — PCR optimization and troubleshooting, experimental design, data interpretation, and scientific writing.

Technical Skills

Molecular Biology: DNA/RNA extraction, PCR, qPCR, RT-qPCR, ddPCR, gel electrophoresis, cloning, nucleic acid purification

Library Prep & Sequencing: Hi-C library preparation, Sanger sequencing, Illumina WGS sample preparation and submission

Cytogenetics & Imaging: Chromosome preparation, FISH probe synthesis and hybridization, fluorescence microscopy (Zeiss ZEN), ImageJ

Genomics & Bioinformatics: Hi-C scaffolding (Juicer, Juicebox), genome assembly from Nanopore long-read data, trio-binning, gene annotation, alignment (BWA, Minimap2)

Population Genetics: FST, PCA, AMOVA, STRUCTURE, PLINK, inversion genotyping

Cell & Assay Work: Mammalian and insect cell culture, sterile technique, ELISA, Western blot, BSL-2 practice

Programming & Software: R (ggplot2), Python, Linux/Unix, MATLAB, Galaxy, Geneious, BLAST/NCBI, GraphPad Prism, BioRender

Scientific Communication: Manuscript preparation, grant proposal writing, poster and oral presentation, mentoring, laboratory documentation

Service and Professional Membership

- Volunteer, Blacksburg Mosquito Control Initiative — field collection and public outreach on vector-borne disease prevention.
- Member, Entomological Society of America.
- Member, Virginia Mosquito Control Association.